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Institution: Algerian Space Agency — Satellite Development Center

Project Title: Predictive Modeling of Li-ion Battery Health and Lifetime for Reliable CubeSat Power Systems

Brief Project Description — Short Summary

This project aims to predict the State of Health (SOH) and Remaining Useful Life (RUL) of lithium-ion batteries using the NASA Battery Dataset. Students will preprocess time-series data from several cells tested until failure, extract meaningful features (voltage, current, temperature, capacity...), and apply data mining and machine learning methods to build predictive models.

The results could serve as a basis for an onboard battery health monitoring system in nanosatellites (CubeSats), enabling preventive maintenance strategies, improving mission reliability, and reducing risks associated with unexpected battery failures.

Brief Project Description — Detailed Description

Reliable power systems are critical for the success of nanosatellite (CubeSat) missions. Lithium-ion batteries are widely used due to their high energy density, but their performance degrades over time. Early prediction of battery degradation and lifetime is essential for mission planning and risk reduction.

This project will use the NASA Battery Dataset, which contains time-series data from several Li-ion cells tested until failure. Students will preprocess and structure the raw data, extract key features (voltage, current, capacity, temperature, internal resistance, etc.), and build predictive models to:

- estimate the State of Health (SOH) of batteries at each cycle,
- predict their Remaining Useful Life (RUL).

The project will explore data mining and machine learning methods (e.g., regression, ensemble models, deep learning) and evaluate their performance on real-world data.

Type of Data:

- Numerical/Tabular (sensor readings)
- Time-series/Sequential (event logs, IoT signals)

Primary Data Mining Method: Regression

Added Value of the Project:

- Practical relevance for space systems: CubeSats and nanosatellites heavily rely on Li-ion batteries. Predicting battery health and lifetime is directly linked to mission success and avoiding costly failures.

- Bridging academia and industry: Students work on real-world data (NASA dataset). Results can feed into applied research at the Algerian Space Agency's Satellite Development Center.
- Technical advancement: Building predictive models (SOH & RUL) strengthens knowledge in time-series mining, regression, and anomaly detection. Students learn both feature engineering and deep learning for sequential data.
- Resource for future projects: The cleaned dataset and extracted features can be reused by future cohorts or research teams. It could form the basis of an onboard health monitoring module in CubeSat missions.
- Innovation and risk reduction: By identifying degradation early, missions can implement preventive strategies instead of reactive fixes. This improves the reliability, longevity, and cost efficiency of nanosatellite power systems.

Supplementary Details or Links:

The dataset used in this project is the NASA Battery Dataset, originally collected by the NASA Ames Prognostics Center of Excellence. It contains detailed time-series data from 4 Li-ion cells (B0005, B0006, B0007, B0018) cycled under controlled conditions until complete failure. Each cycle includes measurements of voltage, current, capacity, and temperature at high frequency, resulting in several thousand charge/discharge cycles and over 100 MB of raw data.

Dataset link: (Kaggle mirror)

<https://www.kaggle.com/datasets/patrickfleith/nasa-battery-dataset/data>